



Probability measure on random variables

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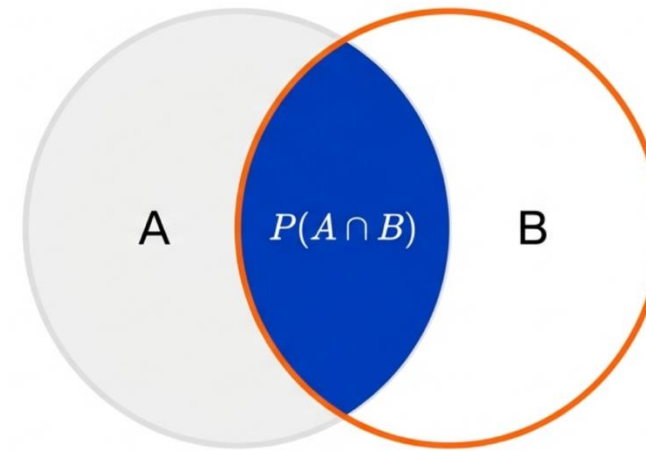


Conditional Probability



Conditional probability measures the occurrence of an event **A**, given that an event **B** has already occurred.

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \text{ con } P(B) > 0$$



- Event **B** becomes the new sample space.
- We no longer care about outcomes outside of **B**; the universe has "shrunk."
- Probability depends on the intersection relative to the size of **B**.



Conditional Probability



Two events are independent if knowing B provides no information about A.

Dependence vs Independence

$$P(A | B) = P(A)$$

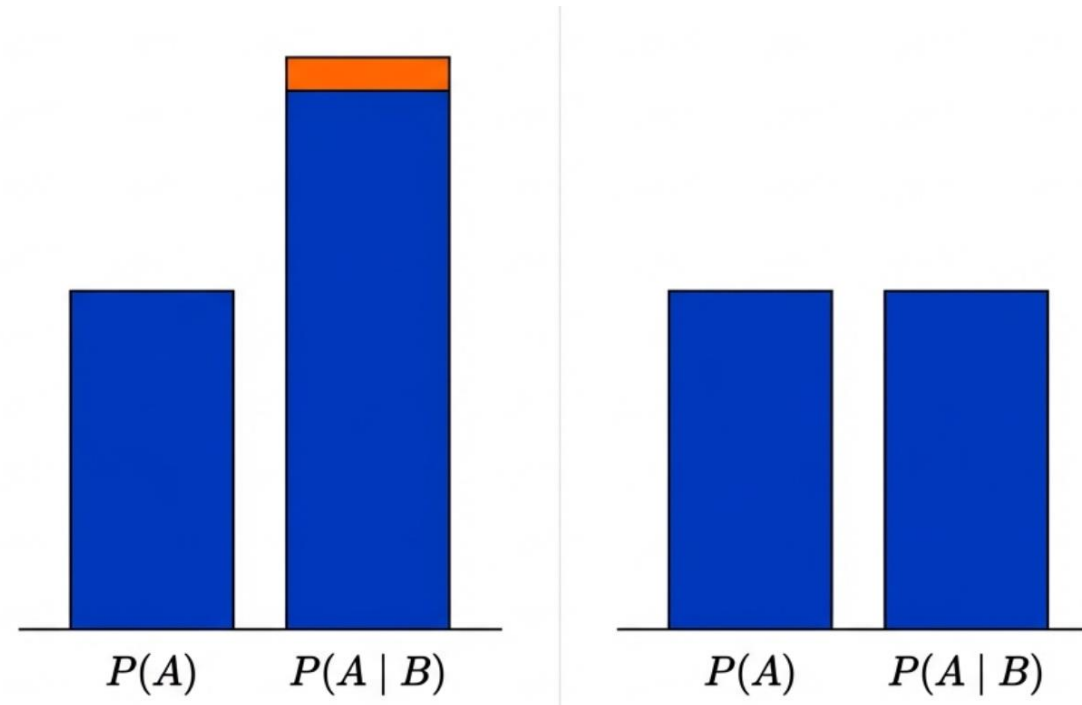
$$P(A \cap B) = P(A) \cdot P(B)$$

Dependence:

The information changes the probability.

Independence:

The information doesn't have an effect





Law of Total Probability

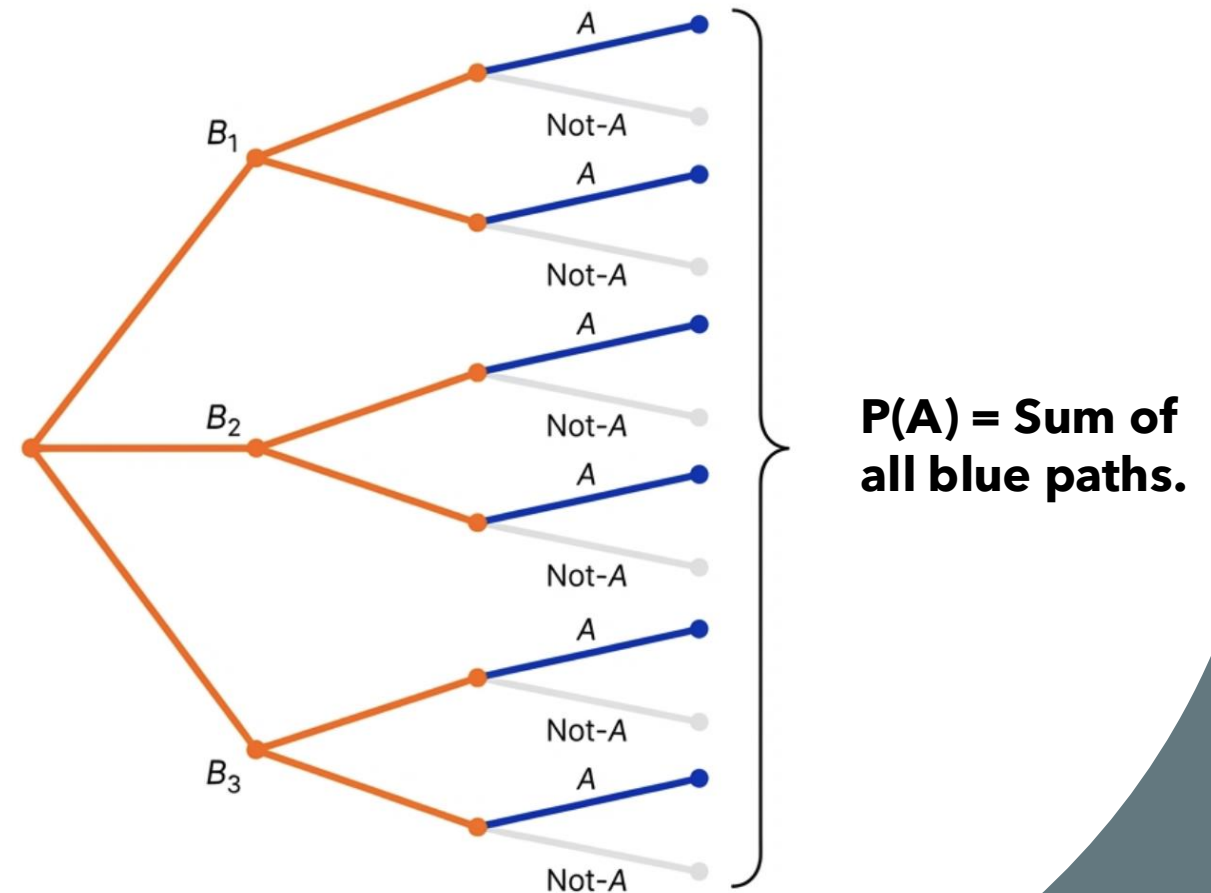


Concept: "Divide and Conquer" Strategy

To calculate a complex event (A), we sum its constituent parts (B_i).

$$P(A) = \sum_{i=1}^n P(A | B_i) \cdot P(B_i)$$

Context: A weighted average of mutually exclusive scenarios.



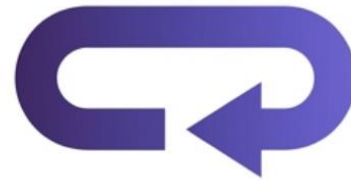


Bayes' Theorem



The mechanism for **updating beliefs** when faced with new evidence.

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$



Prior $P(B)$

Information we had before.



Likelihood $P(A|B)$

The evidence given the cause.



Posterior $P(B|A)$

Updated probability of Cause (B)
after observing Effect (A).



Bayes' Theorem



Intuition vs. Mathematics

Scenario:

- Incidence ($P(A_1)$): 1 out of 1000.
- Precision ($P(B | A_1)$): 99%
- False Positives ($P(B | A_2)$): 2%

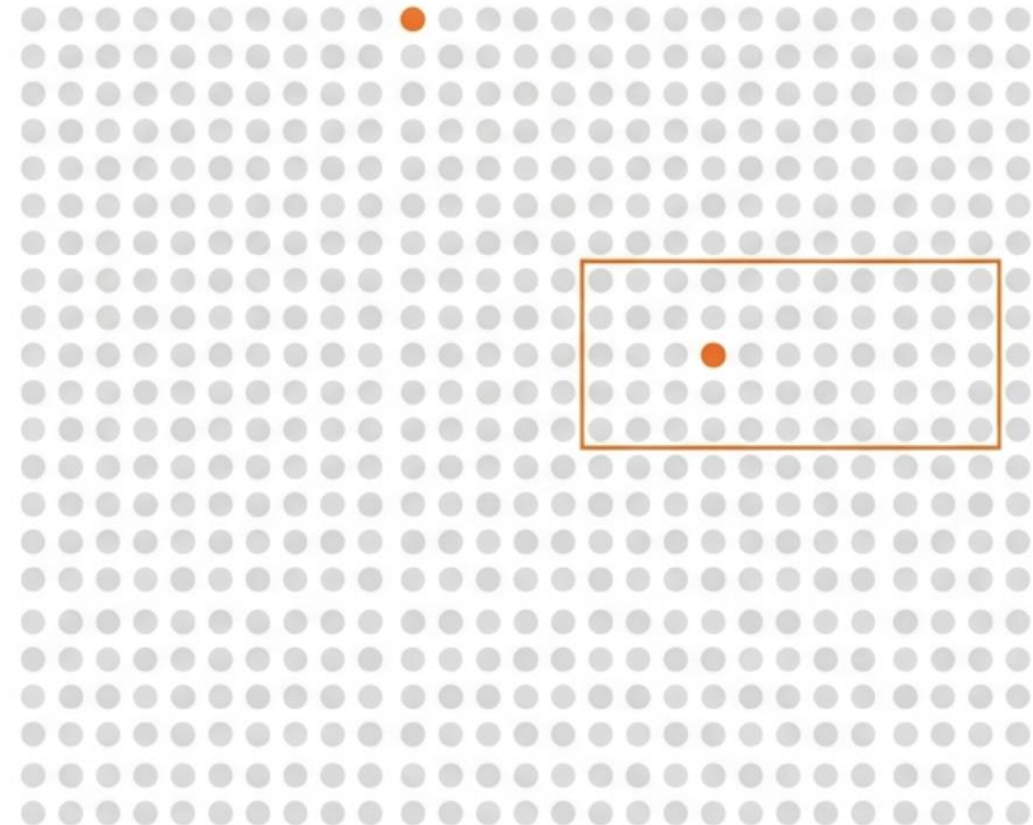
The Question: If you test positive (B), do you have the disease?

Intuition: ~99%

Reality (Bayes): ~4.7%

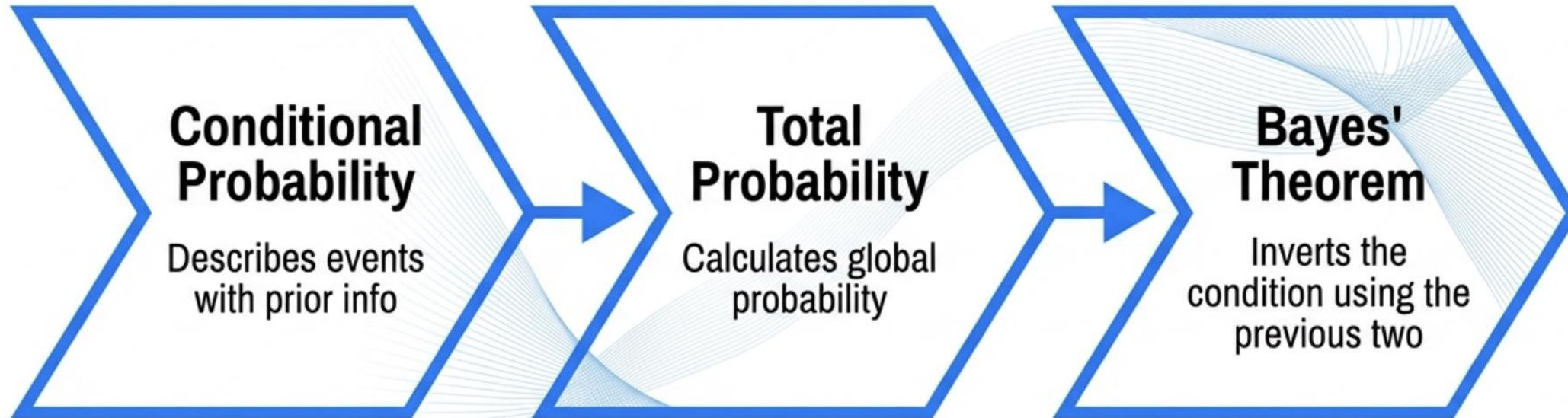
Why? The false positives from the large healthy population “flood” the one real case.

Population Grid





Relationship Between Concepts



$$P(A|B) \rightarrow P(A) \rightarrow P(B|A)$$



References



- Devore, J. L. (2008). Probability and Statistics for Engineering and Science (7th ed.). Cengage Learning.